

Principles and Practices for Building Effective Data Pipelines

Data pipelines ensure information flows smoothly through an organisation, supporting data-driven decisions and analytics. A well-designed pipeline **moves data** from where it is generated to where it is needed, **without interruptions or loss of quality**.

Imagine working on a project that aims to collect and analyse agricultural data to predict crop yields and monitor soil quality to help identify potential food shortages. Effective pipelines would ensure that this data flows seamlessly from farmers, sensors, and suppliers into dashboards where analysts can assess the situation in real-time.

In this train, we'll explore **how to build efficient pipelines** by introducing fundamental principles, discussing ETL vs. ELT architectures, and reviewing tools and strategies for managing costs.

Learning objectives

By the end of this train, you should be able to:

- Define a data pipeline and discuss its role in moving data.
- Describe principles and practices for constructing and maintaining data pipelines.
- Understand how to incorporate modern data pipeline architectures, specifically transitioning from traditional ETL (Extract, Transform, Load) to ELT (Extract, Load, Transform) processes.
- List the common tools available to implement data pipelines.

Key Principles of Building Data Pipelines

A good data pipeline ensures data flows reliably, accurately, and efficiently. Each part must perform smoothly to avoid bottlenecks. There are some essential practices to keep in mind:

Fault tolerance and recovery

Just as a road network should have alternate routes for emergencies, pipelines must recover quickly from disruptions. This could mean retrying tasks automatically or creating backup workflows to prevent data loss.

Data validation

Data that enters the pipeline must be correct and consistent to avoid errors downstream. Imagine a weather station transmitting inconsistent rainfall data—without validation, this could lead to faulty crop predictions.

Modularity and scalability

Pipelines should be built in reusable modules that can scale with the organisation's needs. For example, a data cleaning component can be reused across datasets, whether processing soil moisture readings or crop forecasts.

Automation and monitoring

Automating the pipeline reduces human error and ensures it runs smoothly. A dashboard with real-time alerts helps identify and fix performance issues before they escalate, ensuring the pipeline remains reliable.

ETL vs. ELT: What's the Difference?

The ETL (Extract, Transform, Load) process has been the backbone of data management for years. Here, data is cleaned and transformed before it's stored, ensuring only ready-to-use information goes into the system. However, modern cloud architectures have enabled the rise of ELT (Extract, Load, Transform) pipelines, where raw data is loaded first and processed later inside the data warehouse. Each approach has its strengths:

ETL (Extract, Transform, Load)

Ideal when data requires significant transformation before analysis, such as converting soil sensor readings into a unified format before storing them in a dashboard. This ensures that only clean, structured data enters the system. If a team knows exactly how the data will be used, it makes sense to clean and structure it before storing it.

ELT (Extract, Load, Transform)

Particularly well-suited for raw, unstructured, or unpredictable data that may be used for advanced analysis at a later stage. For example, storing satellite imagery of farmlands over time without knowing upfront which patterns will be significant allows for maximum flexibility. By loading all raw data first, you can apply necessary transformations later when specific analysis requirements or insights become clear, taking full advantage of the data warehouse's processing capabilities.

The choice between ETL and ELT depends on your data needs, infrastructure, and business goals. ETL works best for high-quality, pre-processed data, while ELT offers flexibility and scalability, especially when working with large, unstructured datasets. Organisations transitioning from ETL to ELT can benefit from the cloud's ability to store raw data efficiently, applying transformations only when necessary to meet evolving business needs.

Tools That Support Data Pipelines

Every data pipeline is unique, but some tools have become go-to solutions for building them. Depending on your pipeline's complexity, data type, and performance needs, you might choose different tools to meet specific requirements. Below are some commonly used options:

- **Fivetran:** Automates data extraction and ingestion, ideal for organisations that need to quickly gather data from external sources without writing custom scripts.
- **Apache Kafka:** Best for real-time data ingestion, such as monitoring soil conditions or weather changes, ensuring insights are available the moment the data is collected.
- **Prefect:** A workflow orchestration tool that allows businesses to automate and monitor complex data flows, ensuring that tasks run efficiently across various environments.

Exercise: ETL or ELT Decision-Making

You are building a data pipeline for an agricultural project aimed at reducing food insecurity. The team collects soil data, crop forecasts, and satellite images from multiple sources. Some data needs immediate processing for daily reports, while other datasets will be stored for future research.

Which datasets would you process using ETL, and which using ELT? Explain how the choice of architecture supports the project's goals for both real-time monitoring and long-term planning.

Cost Management in Data Pipelines

Building pipelines in the cloud provides flexibility, but costs can quickly add up without careful management. One of the biggest cost drivers is data storage. To optimise costs, companies often use tiered storage solutions—storing frequently accessed data in high-performance tiers and moving older or less important data to cheaper, slower-access tiers.

Another challenge is data transfer fee. Cloud providers charge for moving data between regions or services, so it's essential to minimise unnecessary movement. For example, storing and processing data within the same cloud region can significantly reduce transfer costs. Additionally, monitoring your pipeline helps identify bottlenecks that may be driving up resource consumption, allowing for timely adjustments.

Monitoring pipeline performance ensures resources are used efficiently. For example, an organisation might detect that it's processing redundant data and adjust the pipeline to save on processing costs.

Exercise: Optimising Pipeline Costs

Your team monitors soil quality across different regions to forecast crop yields. Recently, the cost of transferring data between regions has increased.

Task: Propose a strategy to reduce transfer costs while ensuring data availability for real-time insights.

Here are some ways we thought of. (Try by yourself before looking):

Idea 1

Implement Regional Data Storage: Instead of transferring data across different cloud regions, you can set up regional storage solutions where data collected in a specific region is processed and stored locally. This minimises data transfer costs because the information doesn't need to travel between far-apart locations. By keeping data close to where it is collected and used, you reduce the distance and volume of transfers, which can significantly lower expenses. Furthermore, essential insights can be shared through summarised reports rather than transferring all raw data.

Idea 2

Optimise Data Transfer Frequency and Volume: Instead of transferring every piece of data in real-time, consider prioritising which data points need immediate attention. For example, you could set up a system that only sends critical updates in real-time while less urgent data is batched and transferred at scheduled intervals. This hybrid approach ensures that key insights remain available when needed, but you avoid the constant data transfer costs associated with moving all data continuously.

Conclusion

Building effective data pipelines is about ensuring reliability, scalability, and cost-efficiency. Whether using ETL for structured reports or ELT for raw data exploration, pipelines must be designed with both immediate needs and future flexibility in mind. Thoughtful use of automation, fault tolerance, and monitoring will keep your pipelines running smoothly, helping you generate insights that can drive meaningful impact.

Reflect:

Think of a task you manage in stages, like organising your inbox or planning weekly meals. How could automating parts of this task save time and improve the outcome?